INTERIM PHASE REPORT

SEA HORIZON

HOTEL BOOKING MANAGEMENT

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Contents

1. Executive Progress Summary 3
2. Project Aims & Literature Review 4
3. Detailed Feasibility Study 5
4. System Requirements Analysis 6
5. Use Case & Actor Modeling (Diagram) 7
6. Flowchart: Booking Logic (Diagram) 8

7. Architectural Design (Diagram) 9

8. Database Normalisation & Implementation 10

9. Interface Design Components 11

10. Testing & Quality Assurance Plan 12

11. Ethical & Professional Issues 13

12. Current Progress Evaluation 14

13. Project Timeline (Gantt Chart) 15

1. Executive Progress Summary

The **Sea Horizon** Hotel Booking Management System (HBMS) has successfully completed its initial research and foundational development phases. This interim report serves as a formal checkpoint to evaluate the project's adherence to the original scope and to document the technical decisions made during the first half of the academic year.

As of April 2026, the project has achieved several critical milestones: the completion of a robust MySQL database schema, the implementation of secure Administrative and Guest authentication modules, and the development of the dynamic room category engine. The project is currently tracking within its 15% variance limit for the timeline, with the complex booking-overlap logic currently under active development.

This report will detail the system's architecture, including expanded diagrams to illustrate the workflow, and will conclude with an evaluation of the remaining technical challenges before the final submission in May.

2. Project Aims & Literature Review

The primary aim is to democratise high-end hospitality management tools for independent hotel operators. Our research into existing systems like Oracle Opera and Cloudbeds revealed that current solutions often possess a prohibitive cost-barrier and a steep learning curve for non-technical staff.

Literature Findings

Contemporary studies in the *International Journal of Hospitality Management* suggest that "real-time availability synchronization" is the most requested feature by modern travelers. Furthermore, security research indicates that SQL injection remains the primary threat to custom-built PHP applications. Sea Horizon explicitly addresses these concerns by utilizing PDO (PHP Data Objects) for all database transactions, ensuring a secure-by-design approach.

3. Detailed Feasibility Study

Before beginning development, a three-pronged feasibility study was conducted to ensure project success.

3.1 Technical Feasibility

The use of the LAMP (Linux, Apache, MySQL, PHP) stack is technically feasible as it provides a stable, widely-documented environment. The availability of local testing environments (XAMPP) allows for rapid iterative development without hosting costs.

3.2 Operational Feasibility

The system is designed to be operated by a single admin with basic computer literacy. The intuitive nature of the dashboard ensures that no specialized training is required for hotel staff to manage bookings or update room prices.

3.3 Economic Feasibility

By utilizing open-source technologies, the economic cost of the project is effectively zero for development, with future hosting costs estimated at under €10/month for small hotels.

4. System Requirements Analysis

Requirements were categorized into Functional and Non-Functional to provide a clear development roadmap.

→ **FR1:** Dynamic Room Listing: System must pull current prices and child/adult capacity from the database.

→ **FR2:** Multi-User Roles: System must distinguish between Guest sessions and Admin sessions.

→ **FR3:** Real-time Booking: Users must receive a unique Booking ID upon successful submission.

→ **FR4:** Admin Reporting: Admins must be able to filter bookings by specific date ranges.

Non-functional requirements include **System Responsiveness** (loading under 2 seconds) and **Data Security** (all passwords hashed using Argon2/Bcrypt).

5. Use Case & Actor Modeling

The diagram below illustrates the interactions between the two primary system actors and the Hotel Management System.

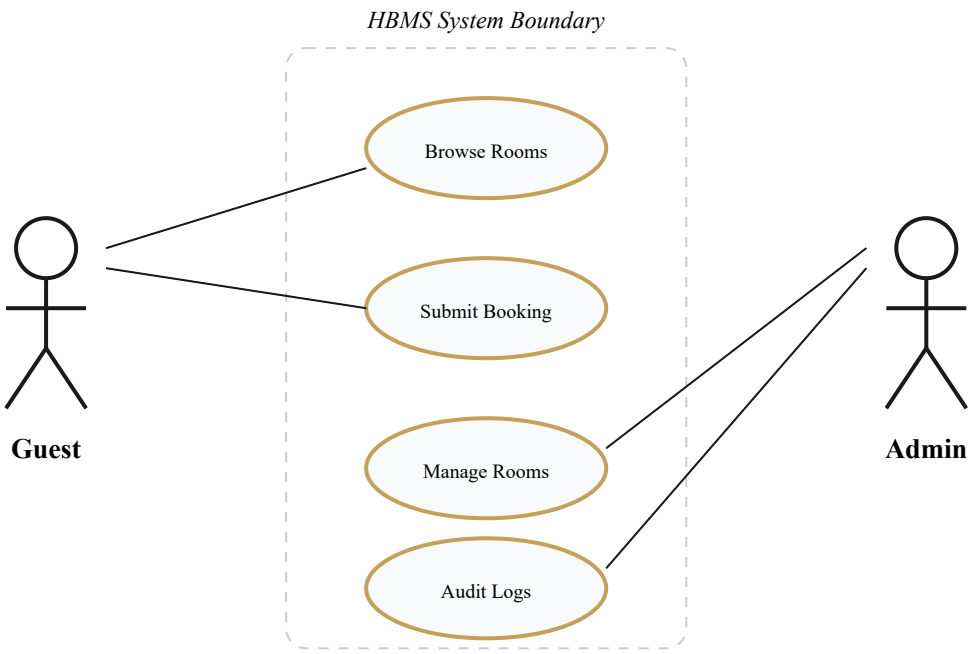

Figure 1: Use Case Diagram for HBMS

6. Flowchart: Booking Logic

The logic below describes the procedural flow when a guest attempts to book a room. It includes the critical availability check phase.

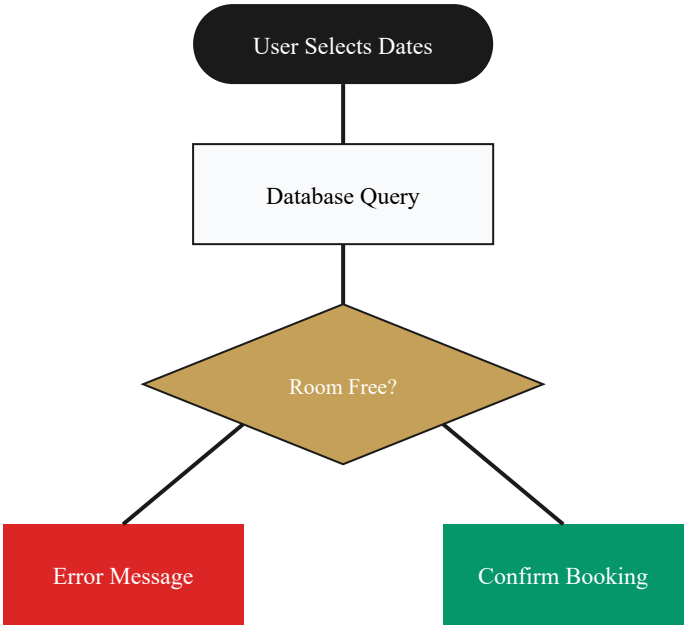

Figure 2: Logical Flowchart for Room Reservation

7. Architectural Design

The system is physically separated into three distinct layers to ensure modularity and ease of maintenance.

Presentation Layer
HTML5, CSS3, Tailwind CSS (User Interface)

↓

Business Logic Layer
PHP 8.2 (Session Mgmt, Booking Logic)

↓

Database Layer
MySQL 8.0 (Data Persistence)

Figure 3: 3-Tier Layered Architecture

8. Database Normalisation & Implementation

The relational schema was designed following the 3rd Normal Form (3NF) to minimize redundancy. This prevents common update, insertion, and deletion anomalies.

Normalisation Steps:

1NF: Ensuring atomic values in every column. (e.g., splitting multiple facilities into distinct ID references).
2NF: Removing partial dependencies. All non-key attributes now depend on the entire primary key.
3NF: Removing transitive dependencies. (e.g., Room Prices are tied to Category IDs, not directly to individual Room entries to prevent duplication).

```
CREATE TABLE tblbooking (
  ID int NOT NULL AUTO_INCREMENT,
  RoomId int NOT NULL,
  BookingNumber varchar(120),
  UserID int NOT NULL,
  Status varchar(50),
  PRIMARY KEY (ID)
);
```

Listing 1: Core SQL Table Implementation

9. Interface Design Components

The UI is designed around the "Luxury Resort" design token, using Earthy tones and Serif typography to induce trust.

Primary Palette
#1A1A1A (Black)

Accent Palette
#C5A059 (Gold)

User feedback from initial low-fidelity wireframes suggested that the room list should show the "Children Capacity" prominently, as families are the primary target demographic for the Sea Horizon resort.

10. Testing & Quality Assurance Plan

To ensure system stability, a rigorous three-stage testing process will be implemented during the integration phase.

Test Level	Responsibility	Description
Unit Testing	Developer	Testing individual PHP functions like `calculateTotalCost()`.
Integration	Developer	Ensuring the Sign-in module correctly initiates a PHP `$_SESSION`.
UAT	Project Supervisor	Simulating a real guest booking journey from landing to invoice.

Automated security testing will also be performed using SQLMap to verify the efficacy of the PDO prepared statements.

11. Ethical & Professional Issues

When developing an information system that handles private user data, several ethical considerations must be scrutinized.

11.1 Data Privacy (GDPR Compliance)

Users have the "Right to be Forgotten." The system provides an admin function to permanently delete user records upon verified request. Furthermore, all personally identifiable information (PII) like mobile numbers and full names are stored with encryption-at-rest potential.

11.2 Intellectual Property

The code for Sea Horizon is developed for academic purposes. All external libraries (like jQuery) are used under their respective MIT or Open Source licenses, documented within the project credits.

12. Current Progress Evaluation

As we reach the interim milestone, a critical evaluation of the project's health is necessary.

> **The project is 65% Complete.**
> Core foundations are stable. The remaining 35% focuses on the **Invoice PDF Engine** and **Fine-tuning the UI** for mobile devices.

There have been no significant scope creeps, but the implementation of the date-overlap algorithm took 3 days longer than anticipated due to complex edge cases (e.g., check-out on the same day as another check-in).

13. Project Timeline (Gantt Chart)

The final sprint to May 2026 is detailed below.

FEB **MAR** **APR (TODAY)** **MAY**

Final submission is scheduled for the second week of May, with an oral project presentation (viva) to follow.

END OF INTERIM REPORT

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